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        "import pandas as pd\n",
        "import numpy as np\n",
        "import plotly.express as px\n",
        "import matplotlib.pyplot as plt\n",
        "import seaborn as sns"
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        "def plot_top_states(df, column_name, chart_title):\n",
        "    # Step 1: Process the data\n",
        "    top_df = df.groupby('State Name')\n",
        "[column_name].sum().sort_values(ascending=False).reset_index().head(10)\n",
        "    \n",
        "    # Step 2: Create the chart\n",
        "    fig = px.bar(top_df, \n",
        "                x='State Name', \n",
        "                y=column_name, \n",
        "                title=chart_title,\n",
        "                template='plotly_white')\n",
        "    \n",
        "    fig.update_layout(height=500, width=800)\n",
        "    return fig.show(renderer=\"iframe\")
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    "# 2. Now, create 'andhra_df' by filtering for the state\\n",

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"andhra_df = df[df['State Name'] == 'Andhra Pradesh']\n",
"\n",
"# 3. Double-check if it worked\n",
"print(f\"✅ 'andhra_df' created successfully with {len(andhra_df)} records.\")"
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"andhra_df = df[df['State Name'] == 'Andhra Pradesh'].copy()\n",
"\n",
"# 2. Filter for the District (Srikakulam)\n",
"srikakulam_df = andhra_df[andhra_df['Dist Name'] == 'Srikakulam'].copy()\n",
"\n",
"# 3. Fact-Check: Did we get anything?\n",
"print(f\"✅ Andhra Pradesh Records: {len(andhra_df)}\")\n",
"print(f\"✅ Srikakulam Records: {len(srikakulam_df)}\")\n",
"\n",
"# Preview the first few years of Srikakulam's data\n",
"srikakulam_df[['Year', 'RICE PRODUCTION (1000 tons)', 'RICE YIELD (Kg per ha)']].head()"
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import plotly.express as px\n",
"\n",
"# We use the srikakulam_df directly\n",
"fig = px.line(\n",
"    srikakulam_df, \n",
"    x='Year', \n",
"    y='RICE YIELD (Kg per ha)',\n",
"    title='50-Year Rice Yield Trend: Srikakulam District',\n",
"    template='plotly_dark'\n",
")\n",
"\n",
"fig.update_layout(\n",
"    xaxis_title='Year (1966-2017)',\n",
"    yaxis_title='Yield (Kg per ha)',\n",
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"\n",
"fig_comp = px.line(\n",
"    comp_df, \n",
"    x='Year', \n",
"    y='RICE YIELD (Kg per ha)', \n",
"    color='Dist Name',\n",
"    title='The Yield Gap: Srikakulam vs East Godavari',\n",
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    "srikakulam_df = andhra_df[andhra_df['Dist Name'] == 'Srikakulam'].copy()\n",
    "east_godavari_df = andhra_df[andhra_df['Dist Name'] == 'East Godavari'].copy()\n",
    "\n",
    "# --- STEP 2: The Comparison Chart ---\n",
    "import plotly.express as px\n",
    "\n",
    "# Combine them for the chart\n",
    "comparison_df = andhra_df[andhra_df['Dist Name'].isin(['Srikakulam', 'East Godavari'])]\n",
    "\n",
    "fig = px.line(\n",
    "    comparison_df, \n",
    "    x='Year', \n",
    "    y='RICE YIELD (Kg per ha)', \n",
    "    color='Dist Name', \n",
    "    title='Productivity Gap: Srikakulam vs East Godavari (1966-2017)', \n",
    "    template='plotly_dark'\n",
    ")\n",
    "\n",
    "fig.update_layout(\n",
    "    xaxis_title='Year', \n",
    "    yaxis_title='Rice Yield (Kg per ha)', \n",
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    "            labels={'State Name':'State Name','Rice Production':'RICE PRODUCTION (1000"
  ],

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tons)'}},\n",
    "        color='RICE PRODUCTION (1000 tons)', color_continuous_scale='Viridis'\n",
    "    )\n",
    "fig.update_layout(xaxis_title='State names', yaxis_title='Rice Production')\n",
    "fig.update_layout(height=600, width=800)\n",
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        "        labels={'State Name':'State Name','Rice Area':'RICE AREA (1000 ha)'}},\n",
        "        color='RICE AREA (1000 ha)',\n",
        "        line_shape=\"linear\",\n",
        "        markers=True\n",
        "    )\n",
        "fig.update_layout(xaxis_title='State names', yaxis_title='Rice Production')\n",
        "fig.update_traces(marker=dict(size=20))\n",
        "fig.update_layout(height=600, width=800)\n",
        "fig.show()"
    ]
},
{
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    "metadata": {},
    "source": [

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```

    "<b><span style=\"color: #3C0753; background-color:#AAD7D9; padding: 20px; font-size: 18px;
border-left: 8px solid #FF9800\">iii) By year</span></b>"
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"rice_year=df.groupby('Year')['RICE PRODUCTION (1000
tons)'].sum().sort_values(ascending=False).reset_index().head(10)\n",
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"shell.execute_reply.started": "2026-05-09T16:36:07.253885Z"
},
"trusted": true
},
"outputs": [],
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"fig = px.pie(rice_year, names='Year', values='RICE PRODUCTION (1000 tons)',\n",
"              title='Total Rice Production by 10 years',\n",
"              labels={'Year': 'Year', 'RICE PRODUCTION (1000 tons)': 'RICE PRODUCTION (1000
tons)'},\n",
"              color='Year', color_discrete_sequence=px.colors.sequential.Viridis,\n",
"              hole=0.1\n",
"              )\n",
"fig.update_layout(height=600,width=800)\n",
"fig.show()"
]
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"<b><span style=\"color: #3C0753; background-color:#AAD7D9; padding: 20px; font-size: 18px;
border-left: 8px solid #FF9800\">iv) Rice yield</span></b>"
]
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"shell.execute_reply.started": "2026-05-09T16:36:07.467647Z"
}
}
}

```

```

    },
    "trusted": true
  },
  "outputs": [],
  "source": [
    "rice_yield=df.groupby('State Name')['RICE YIELD (Kg per
ha)'].sum().sort_values(ascending=False).reset_index().head(10)\n",
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    "rice_yield"
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},
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    },
    "trusted": true
  },
  "outputs": [],
  "source": [
    "fig = px.bar(rice_yield, x='State Name', y='RICE YIELD (Kg per ha)',\n",
    "              title='Total Rice yield by 10 states',\n",
    "              labels={'State Name': 'State Name', 'RICE YIELD ': 'RICE YIELD (Kg per
ha)'}),\n",
    "              color='RICE YIELD (Kg per ha)', color_continuous_scale='inferno'\n",
    "              )\n",
    "\n",
    "fig.update_layout(xaxis_title='State names', yaxis_title='Rice yield')\n",
    "fig.update_layout(height=600,width=800)\n",
    "fig.show()"
  ]
},
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    "    <p style=\"font-size: 40px; font-family: 'Verdana'; color: #3A405A; text-align:
center;\">Wheat</p>\n",
    "</div>"
  ]
},
{
  "cell_type": "markdown",
  "metadata": {},
  "source": [
    "<b><span style=\"color: #3C0753; background-color:#F037A5; padding: 20px; font-size: 18px;
border-left: 8px solid F037A5\">i) By production</span></b>"
  ]
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    "trusted": true
  },
  "outputs": [],

```

```

"outputs": [],
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    "wheat_state=df.groupby('State Name')['WHEAT PRODUCTION (1000
tons)'].sum().sort_values(ascending=False).reset_index().head(10)\n",
    "\n",
    "wheat_state"
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},
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        },
        "trusted": true
    },
    "outputs": [],
    "source": [
        "fig = px.sunburst(wheat_state, path=['State Name'], values='WHEAT PRODUCTION (1000
tons)',\n",
        "                    title='Total Wheat Production by 10 states',\n",
        "                    color='WHEAT PRODUCTION (1000 tons)',\n",
        "                    color_continuous_scale=px.colors.sequential.Inferno,\n",
        "                    \n",
        "                    )\n",
        "\n",
        "fig.update_layout(height=600, width=800)\n",
        "fig.show()"
    ]
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{
    "cell_type": "markdown",
    "metadata": {},
    "source": [
        "<b><span style='color: #3C0753; background-color:#F037A5; padding: 20px; font-size: 18px; border-left: 8px solid F037A5'>ii) By area</span></b>"
    ]
},
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        "trusted": true
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    "outputs": [],
    "source": [
        "wheat_area=df.groupby('State Name')['WHEAT AREA (1000
ha)'].sum().sort_values(ascending=False).reset_index().head(10)\n",
        "\n",
        "wheat_area"
    ]
},
{
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  "trusted": true
},
"outputs": [],
"source": [
    "fig = px.bar(wheat_area, x='State Name', y='WHEAT AREA (1000 ha)',\n",
    "              title='Total Wheat Area by 10 states',\n",
    "              labels={'State Name': 'State Name', 'WHEAT AREA ': 'WHEAT AREA (1000 ha)'},\n",
    "              color='WHEAT AREA (1000 ha)', color_continuous_scale='sunset'\n",
    "              )\n",
    "\n",
    "fig.update_layout(xaxis_title='State names', yaxis_title='Wheat Area')\n",
    "fig.update_layout(height=600,width=800)\n",
    "fig.show()"
]
},
{
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  "metadata": {},
  "source": [
    "<b><span style='color: #3C0753; background-color:#F037A5; padding: 20px; font-size: 18px; border-left: 8px solid F037A5'>iii) By yield</span></b>"
  ]
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    }
  },
  "trusted": true
},
"outputs": [],
"source": [
    "wheat_yield=df.groupby('State Name')['WHEAT YIELD (Kg per ha)'].sum().sort_values(ascending=False).reset_index().head(10)\n",
    "\n",
    "wheat_yield"
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    }
  },
  "trusted": true
},
"outputs": [],
"source": [
    "fig = px.line(wheat_yield, x='State Name', y='WHEAT YIELD (Kg per ha)',\n",
    "              title='Total Wheat yield by 10 states',\n",
    "              labels={'State Name': 'State Name', 'WHEAT yield ': 'WHEAT YIELD (Kg per ha)'},\n",

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        color='WHEAT YIELD (Kg per ha)',\n",
        line_shape="linear",\n",
        markers=True,\n",
        \n",
        )\n",
    "\n",
    "fig.update_layout(xaxis_title='State names', yaxis_title='Wheat Area')\n",
    "fig.update_layout(template='gridon')\n",
    "fig.update_traces(marker=dict(size=30))\n",
    "fig.update_layout(height=600,width=800)\n",
    "fig.show()"
]
},
{
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        "<div style=\"background-color: #F6B17A; #padding: 20px; border-radius: 20px;\">\n",
        "    <p style=\"font-size: 40px; font-family: 'Verdana'; color: #3A405A; text-align: center;\">Barley</p>\n",
        "</div>"
    ]
},
{
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    "metadata": {},
    "source": [
        "<b><span style=\"color: #3C0753; background-color:#F6B17A; padding: 20px; font-size: 18px; border-left: 8px solid F037A5\">i) By area</span></b>"
    ]
},
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        }
    },
    "trusted": true
},
{
    "outputs": [],
    "source": [
        "barley_area=df.groupby('State Name')['BARLEY AREA (1000 ha)'].sum().sort_values(ascending=False).reset_index().head(10)\n",
        "\n",
        "barley_area"
    ]
},
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            "iopub.status.idle": "2026-05-09T16:36:07.893515Z",
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            "shell.execute_reply.started": "2026-05-09T16:36:07.836449Z"
        }
    },
    "trusted": true
},
{
    "outputs": [],
    "source": [
        "fig = px.bar(barley_area, x='State Name', y='BARLEY AREA (1000 ha)',\n",
        "            title='Total Barley Area by 10 states',\n"
    ]

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"          labels={'State Name': 'State Name', 'BARLEY AREA ': 'BARLEY AREA (1000
ha)'}},\n",
"          color='BARLEY AREA (1000 ha)', color_continuous_scale='icefire'\n",
"          )\n",
"\n",
"fig.update_layout(xaxis_title='State names', yaxis_title='Barley Area')\n",
"fig.update_layout(template='gridon')\n",
"fig.update_layout(height=600,width=800)\n",
"fig.show()"
]
},
{
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  "metadata": {},
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border-left: 8px solid F037A5\">ii) By production</span></b>"
  ]
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},
{
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  "source": [
    "barley_pro=df.groupby('State Name')['BARLEY PRODUCTION (1000
tons)'].sum().sort_values(ascending=False).reset_index().head(10)\n",
    "\n",
    "barley_pro"
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    }
  },
  "trusted": true
},
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  "source": [
    "fig = px.scatter(\n",
    "    barley_pro,\n",
    "    x=\"State Name\",\n",
    "    y=\"BARLEY PRODUCTION (1000 tons)\",\n",
    "    size=\"BARLEY PRODUCTION (1000 tons)\",\n",
    "    color=\"BARLEY PRODUCTION (1000 tons)\",\n",
    "    title=\"Barley Production by 10 States\",\n",
    "    color_continuous_scale='Magma',\n",
    "    size_max=40)\n",
    "\n",
    "fig.update_layout(\n",
    "    height=600,\n",
    "    width=800,\n",
    "    xaxis_title=\"State Name\",

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        "yaxis_title=\"Barley Production (1000 tons)\",\n",
        "showlegend=True, # Show legend if using color\n",
        ")\n",
        "fig.update_layout(template='gridon')\n",
        "fig.show()"
    ]
},
{
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    "metadata": {},
    "source": [
        "<b><span style='color: #3C0753; background-color:#F6B17A; padding: 20px; font-size: 18px; border-left: 8px solid F037A5'>iii) By yield</span></b>"
    ]
},
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    },
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    "outputs": [],
    "source": [
        "barley_yield=df.groupby('State Name')['BARLEY YIELD (Kg per ha)'].sum().sort_values(ascending=False).reset_index().head(10)\n",
        "\n",
        "barley_yield"
    ]
},
{
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        }
    },
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    "outputs": [],
    "source": [
        "fig = px.pie(barley_yield, names='State Name', values='BARLEY YIELD (Kg per ha)',\n",
        "            title='Total Barley yield by 10 states',\n",
        "            labels={'State Name': 'State Name', 'BARLEY YIELD (Kg per ha) ': 'BARLEY YIELD (Kg per ha)'},\n",
        "            color_discrete_sequence=px.colors.sequential.Turbo\n",
        "            )\n",
        "\n",
        "fig.update_layout(xaxis_title='State names', yaxis_title='Barley yield')\n",
        "fig.update_layout(height=600,width=800)\n",
        "fig.update_layout(template='gridon')\n",
        "fig.show()"
    ]
},
{
    "cell_type": "markdown",
    "metadata": {},
    "source": [
        "<div style='background-color: #EA9085; #padding: 20px; border-radius: 20px;'>\n",

```

```

"    <p style=\"font-size: 40px; font-family: 'Verdana'; color: #3A405A; text-align:
center;\">Sunflower</p>\n",
"</div>"
]
},
{
"cell_type": "markdown",
"metadata": {},
"source": [
"<b><span style=\"color: #3C0753; background-color:#EA9085; padding: 20px; font-size: 18px;
border-left: 8px solid F037A5\">i) By area</span></b>"
]
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"outputs": [],
"source": [
"sunflower_area=df.groupby('State Name')['SUNFLOWER AREA (1000
ha)'].sum().sort_values(ascending=False).reset_index().head(10)\n",
"\n",
"sunflower_area"
]
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},
"trusted": true
},
"outputs": [],
"source": [
"fig = px.bar(sunflower_area, x='State Name', y='SUNFLOWER AREA (1000 ha)',\n",
"            title='Total Sunflower Area by 10 states',\n",
"            labels={'State Name': 'State Name', 'SUNFLOWER AREA (1000 ha) ': 'SUNFLOWER AREA
(1000 ha)'},\n",
"            color='SUNFLOWER AREA (1000 ha)', color_continuous_scale='viridis'\n",
"            )\n",
"\n",
"fig.update_layout(xaxis_title='State names', yaxis_title='Sunflower Area')\n",
"\n",
"fig.update_layout(template='plotly_white')\n",
"fig.update_layout(height=600,width=800)\n",
"fig.show()"
]
},
{
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"metadata": {},
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"<b><span style=\"color: #3C0753; background-color:#EA9085; padding: 20px; font-size: 18px;
border-left: 8px solid F037A5\">ii) By production</span></b>"
]
}

```

```

    ],
    },
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        "    x=\"State Name\",\n",
        "    y=\"SUNFLOWER PRODUCTION (1000 tons)\",\n",
        "    size=\"SUNFLOWER PRODUCTION (1000 tons)\",\n",
        "    color=\"SUNFLOWER PRODUCTION (1000 tons)\",\n",
        "    title=\"Sunflower Production by 10 States\",\n",
        "    color_continuous_scale='Inferno',\n",
        "    size_max=70)\n",
        "\n",
        "fig.update_layout(\n",
        "    height=600,\n",
        "    width=800,\n",
        "    xaxis_title=\"State Name\",\n",
        "    yaxis_title=\"SUNFLOWER PRODUCTION (1000 tons)\",\n",
        "    showlegend=True,\n",
        ")\n",
        "fig.update_layout(template='plotly_white')\n",
        "fig.show()"
      ]
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  "sun_yield"
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    "              color='State Name',color_discrete_sequence=px.colors.sequential.Burgyl\n",
    "              )\n",
    "\n",
    "fig.update_layout(height=600, width=800)\n",
    "fig.update_layout(template='plotly_white')\n",
    "fig.show()"
  ]
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    "  <p style=\"font-size: 40px; font-family: 'Verdana'; color: #3A405A; text-align:\n",
    "center;\">Soyabean</p>\n",
    "</div>"
  ]
},
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    "border-left: 8px solid F037A5\">i) By area</span></b>"
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    "              title='Total Soyabean Area by 10 states',\n",
    "              labels={'State Name': 'State Name', 'SOYABEAN AREA (1000 ha)': 'SOYABEAN AREA (1000 ha)'},\n",
    "              color='SOYABEAN AREA (1000 ha)', color_continuous_scale='purpor'\n",
    "              )\n",
    "\n",
    "fig.update_layout(xaxis_title='State names', yaxis_title='Soyabean Area')\n",
    "\n",
    "fig.update_layout(template='ggplot2')\n",
    "fig.update_layout(height=600,width=800)\n",
    "fig.show()"
]
},
{
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    "<b><span style='color: #3C0753; background-color:#E0AED0; padding: 20px; font-size: 18px; border-left: 8px solid F037A5'>ii) By production</span></b>"
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    }
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```

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]
},
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  "source": [
    "fig = px.line(soya_pro, x='State Name', y='SOYABEAN PRODUCTION (1000 tons)',\n",
    "               title='Total Soyabean yield by 10 states',\n",
    "               labels={'State Name': 'State Name', 'SOYABEAN PRODUCTION (1000 tons)':\n",
    "'SOYABEAN PRODUCTION (1000 tons)'},\n",
    "               color='SOYABEAN PRODUCTION (1000 tons)',\n",
    "               line_shape='linear',\n",
    "               markers=True)\n",
    "\n",
    "fig.update_layout(xaxis_title='State names', yaxis_title='Soya Area')\n",
    "fig.update_layout(template='ggplot2')\n",
    "fig.update_layout(height=600,width=800)\n",
    "fig.update_traces(marker=dict(size=20))\n",
    "fig.show()"
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  "outputs": [],
  "source": [
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    "\n",
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"outputs": [],
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    "              title='Total Soyabean yield by 10 states',\n",
    "              labels={'Year': 'Year', 'SOYABEAN YIELD (Kg per ha)': 'SOYABEAN YIELD (Kg per\n",
    ha)'}),\n",
    "              color='SOYABEAN YIELD (Kg per ha)',\n",
    color_discrete_sequence=px.colors.sequential.Oranges ,\n",
    "              hole=0.05\n",
    "              )\n",
    "\n",
    "fig.update_layout(template='ggplot2')\n",
    "fig.update_layout(height=600,width=800)\n",
    "fig.show()"
  ]
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    "</div>"
  ]
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    "\n",
    "oil_area"
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  "              title='Total Oilseeds Area by 10 states',\n",
  "              labels={'State Name': 'State Name', 'OILSEEDS AREA (1000 ha) ': 'OILSEEDS AREA\n",
  "(1000 ha)'}),\n",
  "              color='OILSEEDS AREA (1000 ha)', color_continuous_scale='blackbody'\n",
  "              )\n",
  "\n",
  "fig.update_layout(xaxis_title='State names', yaxis_title='Soyabean Area')\n",
  "\n",
  "fig.update_layout(template='ggplot2')\n",
  "fig.update_layout(height=600,width=800)\n",
  "fig.show()"
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  "\n",
  "oil_pro"
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  },
  "trusted": true
},
"outputs": [],
"source": [
  "fig = px.scatter(\n",
  "    oil_pro,\n",

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```

"    x=\"State Name\", \"n\",
"    y=\"OILSEEDS PRODUCTION (1000 tons)\", \"n\",
"    size=\"OILSEEDS PRODUCTION (1000 tons)\", \"n\",
"    color=\"OILSEEDS PRODUCTION (1000 tons)\", \"n\",
"    title=\"Oilseed Production by 10 States\", \"n\",
"    color_continuous_scale='Sunset', \"n\",
"    size_max=40)\", \"n\",
"\"n\",
"fig.update_layout(\"n\",
"    height=600, \"n\",
"    width=800, \"n\",
"    xaxis_title=\"State Name\", \"n\",
"    yaxis_title=\"Oilseed production (1000 tons)\", \"n\",
"    showlegend=True, \"n\",
")\"n\",
"fig.update_layout(template='ggplot2')\"n\",
"fig.show()"
]
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},
"outputs": [],
"source": [
"fig = px.pie(oil_yield, names='State Name', values='OILSEEDS YIELD (Kg per ha)', \"n\",
"    title='Total Soyabean yield by 10 states', \"n\",
"    labels={'State Name': 'State Name', 'OILSEEDS YIELD (Kg per ha)': 'OILSEEDS
YIELD (Kg per ha)'}, \"n\",
"    color='State Name', color_discrete_sequence=px.colors.sequential.Blackbody , \"n\",
"    hole=0.1\"n\",

```

```

        "\n",
        "fig.update_layout(template='ggplot2')\n",
        "fig.update_layout(height=600,width=800)\n",
        "fig.show()"
    ]
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        "</div>"
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        "<b><span style=\"color: #3C0753; background-color:#CD8D7B; padding: 20px; font-size: 18px; border-left: 8px solid F037A5\">i) By area</span></b>"
    ]
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},
{
    "outputs": [],
    "source": [
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        "\n",
        "cotton_area"
    ]
},
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},
{
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    "source": [
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        "              title='Total Cotton Area by Top 10 states',\n",
        "              labels={'State Name': 'State Name', 'COTTON AREA (1000 ha)': 'COTTON AREA (1000 ha)'},\n",
        "              color='COTTON AREA (1000 ha)', color_continuous_scale='inferno'\n",
        "              )\n",
        "\n",
        "fig.update_layout(xaxis_title='State names', yaxis_title='Cotton Area')\n",

```

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    "\n",
    "fig.update_layout(template='simple_white')\n",
    "fig.update_layout(height=600,width=800)\n",
    "fig.show()"
]
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    "<b><span style=\"color: #3C0753; background-color:#CD8D7B; padding: 20px; font-size: 18px; border-left: 8px solid F037A5\">ii) By production</span></b>"
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    "cot_pro"
  ]
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    "    color=\"COTTON PRODUCTION (1000 tons)\",\n",
    "    title=\"Cotton Production by top 10 States\",\n",
    "    color_continuous_scale='Blackbody',\n",
    "    size_max=40)\n",
    "\n",
    "fig.update_layout(\n",
    "    height=600,\n",
    "    width=800,\n",
    "    xaxis_title=\"State Name\",\n",
    "    yaxis_title=\"Cotton production (1000 tons)\",\n",
    "    showlegend=True,\n",
    ")\n",
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}

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        "                  title='Total Cotton Yield by top 10 states',\n",
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        "                  color_continuous_scale=px.colors.sequential.Blackbody,\n",
        "                  \n",
        "                  )\n",
        "\n",
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        "fig.show()"
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```

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    "              labels={'State Name': 'State Name', 'FRUITS AREA (1000 ha)': 'FRUITS AREA (1000 ha)'},\n",
    "              color='FRUITS AREA (1000 ha)', color_continuous_scale='rainbow'\n",
    "              )\n",
    "\n",
    "fig.update_layout(xaxis_title='State names', yaxis_title='Fruit Area')\n",
    "\n",
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```

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"              color='VEGETABLES AREA (1000 ha)',
color_discrete_sequence=px.colors.sequential.Agsunset\n",
"              \n",
"              )\n",
"\n",
"fig.update_layout(height=600, width=800)\n",
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    "               labels={'State Name': 'State Name', 'FRUITS AND VEGETABLES AREA (1000 ha) ':  

    'FRUITS AND VEGETABLES AREA (1000 ha)'},\n",
    "               color='FRUITS AND VEGETABLES AREA (1000 ha)',\n",
    "               line_shape='linear',\n",
    "               markers=True,\n",
    "               )\n",
    "\n",
    "fig.update_layout(xaxis_title='State names', yaxis_title='Fruits and Vegetables Area')\n",
    "fig.update_layout(template='plotly_dark')\n",
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      "              color='POTATOES AREA (1000 ha)', color_continuous_scale='haline',\n",
      "              )\n",
      "\n",
      "fig.update_layout(xaxis_title='State names', yaxis_title='Potatoes Area')\n",
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      "fig.update_layout(template='plotly_white')\n",
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ha)'},\n",
    "              color='ONION AREA (1000 ha)',
color_discrete_sequence=px.colors.sequential.Magenta ,\n",
    "              )\n",
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center;\">Fodder</p>\n",
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```

"Fodder includes hay, straw, silage, compressed and pelleted feeds, oils and mixed rations, and sprouted grains and legumes (such as bean sprouts, fresh malt, or spent malt). Most animal feed is from plants, but some manufacturers add ingredients to processed feeds that are of animal origin."

```

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    "    color=\"FODDER AREA (1000 ha)\",\n",
    "    title=\"Fodder Area by top 10 States\",\n",
    "    size_max=50,\n",
    "    color_continuous_scale='Tealgrn')\n",
    "\n",
    "fig.update_layout(\n",
    "    height=600,\n",
    "    width=800,\n",
    "    xaxis_title=\"State Name\",\n",
    "    yaxis_title=\"Fodder areas\",\n",
    "    showlegend=True,\n",
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```

```

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      "              color='KHARIF SORGHUM AREA (1000 ha)',\n",
      "              )\n",
      "\n",
      "fig.update_layout(xaxis_title='State names', yaxis_title='Kharif Sorghum Area')\n",
      "\n",
      "fig.update_layout(template='seaborn')\n",
      "fig.update_layout(height=600,width=800)\n",
      "fig.show()"
    ]
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        "              color='KHARIF SORGHUM PRODUCTION (1000 tons)',\n",
        "              hole=0.1\n",
        "              )\n",
        "\n",
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    "    size=\"KHARIF SORGHUM YIELD (Kg per ha)\",\n",
    "    color=\"KHARIF SORGHUM YIELD (Kg per ha)\",\n",
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    "fig.update_layout(\n",
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    "    width=800,\n",
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    "    yaxis_title=\"Kharif Sorghum yield\",\n",
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SORGHUM AREA (1000 ha)'},\\n\",
    \"          color='RABI SORGHUM AREA (1000 ha)', color_continuous_scale='magma'\\n\",
    \"          )\\n\",
    \"\\n\",
    \"fig.update_layout(xaxis_title='State names', yaxis_title='Rabi Sorghum Area')\\n\",
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    \"fig.show()\"
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        "    color=\"RABI SORGHUM YIELD (Kg per ha)\",\n",
        "    title=\"Rabi Sorghum Yield by top 10 states\",\n",
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        "    color_continuous_scale='Sunset')\n",
        "\n",
        "fig.update_layout(\n",
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        "    width=800,\n",
        "    xaxis_title=\"State Name\",\n",
        "    yaxis_title=\"Rabi Sorghum yield\",\n",
        "    showlegend=True,\n",
        ")\n",
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        "                labels={'State Name': 'State Name', 'MAIZE AREA (1000 ha) ': 'MAIZE AREA (1000
ha)'}\n",
        "                color='MAIZE AREA (1000 ha)', color_continuous_scale='sunset'\n",
        "                )\n",
        "\n",
        "fig.update_layout(xaxis_title='State names', yaxis_title='Maize Area')\n",
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  "            hole=0.1\n",
  "            #color_discrete_sequence=px.colors.sequential.Sunset\n",
  "            )\n",
  "\n",
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  "fig.update_layout(height=600,width=800)\n",
  "fig.show()"
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```

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  "    y=\"MAIZE YIELD (Kg per ha)\",\n",
  "    size=\"MAIZE YIELD (Kg per ha)\",\n",
  "    color=\"MAIZE YIELD (Kg per ha)\",\n",
  "    title=\"MAIZE YIELD (Kg per ha)\",\n",
  "    size_max=30,\n",
  "    color_continuous_scale='Viridis' # Specify the color scale\n",
  ")\n",
  "\n",
  "fig.update_layout(\n",
  "    height=600,\n",
  "    width=800,\n",
  "    xaxis_title=\"State Name\",\n",
  "    yaxis_title=\"Maize yield\",\n",
  "    showlegend=True,\n",
  ")\n",
  "fig.update_layout(template='plotly_dark')\n",
  "fig.show()\n",
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    "        >Finger Millet</p>\n",
    "    </div>\n",
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  "outputs": [],
  "source": [
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    "              labels={'State Name': 'State Name', 'FINGER MILLET AREA (1000 ha) ': 'FINGER\n",
    "MILLET AREA (1000 ha)'},\n",
    "              color='FINGER MILLET AREA (1000 ha)', color_continuous_scale='blackbody'\n",
    "              )\n",
    "\n",
    "fig.update_layout(xaxis_title='State names', yaxis_title='Finger Millet Area')\n",
    "\n",
    "fig.update_layout(template='seaborn')\n",
    "fig.update_layout(height=600,width=800)\n",
    "fig.show()"
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    "border-left: 8px solid F037A5\n",
    ">ii) By production</span></b>"
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  },
  "trusted": true
},
"outputs": [],
"source": [
    "fig = px.pie(fg_pro, names='State Name', values='FINGER MILLET PRODUCTION (1000 tons)',\n",

```

```

"                title='Total Finger Millet production by top 10 states',\n",
"                labels={'State Name': 'State Name', 'FINGER MILLET PRODUCTION (1000 tons)':
'FINGER MILLET PRODUCTION (1000 tons)'},\n",
"                color='FINGER MILLET PRODUCTION (1000 tons)',\n",
"                hole=0.1,\n",
"                color_discrete_sequence=px.colors.sequential.Plasma\n",
"            )\n",
"\n",
"fig.update_layout(template='seaborn')\n",
"fig.update_layout(height=600,width=800)\n",
"fig.show()"
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border-left: 8px solid F037A5\">iii) By yield</span></b>"
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"fg_yi=df.groupby('State Name')['FINGER MILLET YIELD (Kg per
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"fg_yi"
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"    x=\"State Name\",\n",
"    y=\"FINGER MILLET YIELD (Kg per ha)\",\n",
"    size=\"FINGER MILLET YIELD (Kg per ha)\",\n",
"    color=\"FINGER MILLET YIELD (Kg per ha)\",\n",
"    title=\"Finger Millet yield by top 10 states\",\n",
"    size_max=30,\n",
"    color_continuous_scale='Magma' \n",
")\n",
"\n",
"fig.update_layout(\n",

```

```

"    height=600,\n",
"    width=800,\n",
"    xaxis_title=\"State Name\",\n",
"    yaxis_title=\"Finger Millet yield\",\n",
"    showlegend=True,\n",
")\n",
"fig.update_layout(template='seaborn')\n",
"fig.show()"
]
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">Safflower</p>\n",
"</div>"
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"saff_ar=df.groupby('State Name')['SAFFLOWER AREA (1000 ha)'].sum().sort_values(ascending=False).reset_index().head(10)\n",
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"saff_ar"
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"    title='Total Safflower Area by Top 10 states',\n",
"    labels={'State Name': 'State Name', 'SAFFLOWER AREA (1000 ha) ': 'SAFFLOWER AREA (1000 ha)'},\n",
"    color='SAFFLOWER AREA (1000 ha)', color_continuous_scale='aggrnyl'\n",

```

```

        "\n",
        "\n",
        "fig.update_layout(xaxis_title='State names', yaxis_title='Safflower Area')\n",
        "\n",
        "fig.update_layout(template='simple_white')\n",
        "fig.update_layout(height=600,width=800)\n",
        "fig.show()"
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        "<b><span style=\"color: #3C0753; background-color:#9BB8CD; padding: 20px; font-size: 18px; border-left: 8px solid F037A5\">ii) By production</span></b>"
    ]
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        "    values='SAFFLOWER PRODUCTION (1000 tons)',\n",
        "    title='Total Safflower production by top 10 states',\n",
        "    labels={'State Name': 'State Name', 'SAFFLOWER PRODUCTION (1000 tons)': 'SAFFLOWER PRODUCTION (1000 tons)'},\n",
        "    color='SAFFLOWER PRODUCTION (1000 tons)',\n",
        ")\n",
        "# Manually adjust the pull to make all slices fully visible\n",
        "fig.update_traces(pull=[0] * len(saff_pro), textinfo='value+percent',\n        textposition='inside')\n",
        "\n",
        "fig.update_layout(template='simple_white')\n",
        "fig.update_layout(height=600, width=800)\n",
        "fig.show()"
    ]
}

```

```

    ]
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  "    saff_yi,\n",
  "    x='State Name',\n",
  "    y='SAFFLOWER YIELD (Kg per ha)',\n",
  "    size='SAFFLOWER YIELD (Kg per ha)',\n",
  "    color='SAFFLOWER YIELD (Kg per ha)',\n",
  "    title='Safflower yield by top 10 states',\n",
  "    size_max=30,\n",
  "    color_continuous_scale='Tealgrn' \n",
  ")\n",
  "\n",
  "fig.update_layout(\n",
  "    height=600,\n",
  "    width=800,\n",
  "    xaxis_title='State Name',\n",
  "    yaxis_title='Finger Millet yield',\n",
  "    showlegend=True,\n",
  ")\n",
  "fig.update_layout(template='simple_white')\n",
  "fig.show()"
]

```

```

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    "<b><span style=\"color: #3C0753; background-color:#FFF8C9; padding: 20px; font-size: 18px; border-left: 8px solid F037A5\">i) By area</span></b>"
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    "              labels={'State Name': 'State Name', 'SUGARCANE AREA (1000 ha)': 'SUGARCANE AREA (1000 ha)'},\n",
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    "              labels={'State Name': 'State Name', 'SUGARCANE PRODUCTION (1000 tons)':\n",
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    "              color='SUGARCANE PRODUCTION (1000 tons)',\n",
    "              hole=0.1,\n",
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    "              )\n",
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    "fig.show()"
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  "    y=\"SUGARCANE YIELD (Kg per ha)\",\n",
  "    size=\"SUGARCANE YIELD (Kg per ha)\",\n",
  "    color=\"SUGARCANE YIELD (Kg per ha)\",\n",
  "    title=\"Sugarcane yield by top 10 states\",\n",
  "    size_max=30,\n",
  "    \n",
  ")\n",
  "\n",
  "fig.update_layout(\n",
  "    height=600,\n",
  "    width=800,\n",
  "    xaxis_title=\"State Name\",\n",
  "    yaxis_title=\"Finger Millet yield\",\n",
  "    showlegend=True,\n",
  ")\n",
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  "fig.show()"
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    "        Thank you for exploring my notebook!</p>\n",
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